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END SEMESTER EXAMINATION DEC - 2024

Name of the Course: B.Tech

Semester: I

Name of the Paper: Fundamental of Computer and Introduction to Programming

Paper Code: **TCS101**

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) **Q2. is having only two parts. Attempt both the parts.**
- (iv) Total marks in each main question is twenty.
- (v) Each parts carries 10 marks

Q1		(20marks)	CO1
(a)	What is computer? Explain block diagram of computer with its components. Also explain its characteristics.		
(b)	What is the term Computer Network means? Considering size as a parameter explain different kind of networks. Also explain the term topology and its types.		
(c)	Write brief notes about the following terms: 1. System software and Application software 2. GUI and CUI 3. RAM and ROM 4. Linker and Loader		
Q2		(20 marks)	CO2 CO3
(a)		Find output of following codes:	
		1. <pre>int main() { int x=10; for (;x;) x--; printf("%d\n",x); return 0; }</pre> 2. <pre>int main() { int x; float y; y = x = 8.0/3; printf("%d %f", x, y); return 0; }</pre>	
		3. <pre>void check() { int ch = 2; switch(ch) { case 1: printf("1 "); case 2: printf("2 "); case 3: printf("3 "); } } int main() { check(); check(); return 0; }</pre> 4. <pre>void calculate(int x) { if(x == 0) { printf("%d ", x); return; } printf("%d ", x); calculate(x - 1); printf("%d ", x); } int main() { calculate(5); return 0; }</pre>	

	<pre> 5. int main() { int a = 10; do { printf(" %d", a); a++; if(a>15) break; }while(1); } </pre>	
(b)	Draw a flowchart to print following pattern: <pre> ** *** ***** ***** </pre> Till n lines.	
Q3	(20 marks)	
(a)	Explain the various looping construct available in 'C' with proper syntax and example. Also differentiate entry-controlled and exit-controlled loops.	
(b)	Ms. Priya has to solve a problem. She has to read a month number (between 1 to 12) and the output should come by displaying number of days in inputted month. Write a 'C' program to help her to solve the problem using switch-case. Sample Input: Input month number: 4 Sample Output: Number of days: 30 Sample Input: Input month number: 2 Sample Output: Number of days: 28 Sample Input: Input month number: 3 Sample Output: Number of days: 31	CO4
(c)	Mr. X has to check n numbers. He wants to keep the numbers having more than 2 factors. Help Mr. X to find the numbers having more than 2 factors. Write a 'C' program, so that Mr. X can finally show the total count of numbers among n having factors more than 2.	
Q4	(20 marks)	
(a)	Describe function along with its advantages. What do you mean by Call by Value? Explain with an example. Also describe actual parameters and formal parameters.	
(b)	Write a 'C' program to find sum of following series: $1^1/1! + 2^2/2! + 3^3/3! + \dots + n^n/n!$ Create two separate user define functions <i>fact()</i> and <i>power()</i> to calculate factorial and simulate pow().	CO5

(c)	What is recursion? Create a recursive function in 'C' to find sum of digit of an inputted number.	
Q5	(20 marks)	
(a)	What is the term Static Memory Allocation mean? Explain various merits and demerits of the array. Also describe the methods to initialize an array.	CO6
(b)	A certain non-government organization is studying on the nutritional aspects on human height. For the study, the NGO requires to have height (in cms) data of 'N' individuals. The study involves by collecting the data of 'N' individuals and finding the minimum, maximum and average height of the individuals' data. Write a 'C' program to collect the data of 'N' individuals on behalf of the NGO and display the minimum, maximum and average height (using array).	
(c)	'N' kids with specific given ID are standing in a line. A position is called and the kid placed in that position get out from that line and next kid fill that position and a new line is formed. Write a 'C' program show the concept of removing kid from the line and display the final line with their id. Sample Input: Value of N: 7 Kids in line(original): 10 20 30 40 50 60 70 Position number: 4 Sample Output: Kids in line(after removal): 10 20 30 50 60 70	

